

Q1a) Differentiate between **toolkits** and **User Interface Management Systems (UIMS)** on the basis of flexibility and level of systems they support (high/low) only? **[5 marks]**

A1a)

Toolkits

UIMS

Low-level support

High-level support

More flexible

More structured

Q1b) A bank develops an advanced mobile banking application for pensioners that has been thoroughly tested by the engineering team. The application performs all functions correctly, including money transfer, balance inquiry, and bill payment. No software defects are found during testing.

Which discipline, Software Engineering or Human-Computer Interaction, is better suited to address application's usability and why? **[5 marks]**

A1b) Usability focuses on how easily and effectively users can interact with the system. Since the application is designed for pensioners, factors such as:

- Ease of learning
- Readability of text
- Button size and placement
- Navigation simplicity
- Memory and cognitive load
- User satisfaction

become important. These aspects are studied within Human-Computer Interaction (HCI), which considers human psychology, cognition, perception, and user behavior when designing interfaces.

Q2a) A team is evaluating two versions of an online learning platform — one with a traditional menu and one with a conversational chatbot interface. Explain why a *between-subjects design* would be more appropriate than a within-subjects design for this evaluation. In your answer, describe the experimental procedure and identify which variables must be controlled. **[5 marks]**

A2a) (2 pt) Why between-subjects: Using the learning platform teaches users how to navigate it. If participants use both versions, knowledge gained from the first interface (e.g. learning platform content, navigation conventions) would transfer and inflate performance on the second

— an irreversible carry-over. Between-subjects avoids this: Group 1 uses the menu interface only; Group 2 uses the chatbot only.

(1pt) Procedure: Participants are randomly assigned to one of the two conditions. Both groups complete the same set of learning tasks. Task completion time, accuracy, and satisfaction ratings are measured.

(2pt) Variables to control: Prior experience with similar platforms, age, educational background, and device used — these should be matched or randomised across groups to ensure observed differences reflect the interface, not participant characteristics.

Q2b) A software company has developed a new mobile application for online ticket booking. Before involving real users, the development team wants to evaluate the interface. **[5 marks]**

The team has the following concerns:

- Whether first-time users can easily understand how to book a ticket.
- Whether the interface follows established usability principles.
- Whether experts can compare the design with existing usability guidelines and literature.
- Whether user performance can be predicted mathematically without conducting user tests.

Based on the classification of Analytic Evaluation Techniques, identify the most appropriate technique for each concern and justify your answer.

A2b) 1.

Technique: Cognitive Walkthrough

Justification:

Cognitive Walkthrough focuses on evaluating how easily a new user can learn and perform tasks by examining each step of the interaction process.

2.

Technique: Heuristic Evaluation

Justification:

Heuristic Evaluation uses usability principles (heuristics) such as consistency, feedback, and error prevention to identify usability problems.

3. Technique: Review-Based Evaluation

Justification:

Review-Based Evaluation relies on standards, design guidelines, previous research findings, and usability literature to assess the interface.

4. Predicting User Performance Without User Testing

Technique: Model-Based Evaluation

Justification:

Model-Based Evaluation uses predictive models (such as GOMS or the Keystroke-Level Model) to estimate user performance and task completion time.

Q3) A software company in 1960 developed a computer system for city planners. The system produces large paper reports containing maps, traffic statistics, and infrastructure data. City planners complain that it takes too much time to understand the information and make decisions.

In 1963, a new engineer proposes replacing paper reports with a display-screen system inspired by Ivan Sutherland's Sketchpad

[5+5 (marks)]

Questions:

- a. What contribution of Sketchpad was established regarding the Video Display Unit only?
- b. What is the second important idea that was generated by this invention of Sutherland?

A3a) Sketchpad allowed a computer operator to use the computer to create, very rapidly, sophisticated visual models on a display screen that resembled a television set. The visual patterns could be stored in the computer's memory like any other data, and could be manipulated by the computer's processor.... But Sketchpad was much more than a tool for creating visual displays. It was a kind of simulation language that enabled computers to translate abstractions into perceptually concrete forms.

A3b) Sutherland's efforts demonstrated how important the contribution of one creative mind (coupled with a dogged determination to see the idea through) could be to the entire history of computing.

Q4) A university is developing a **Student Information System (SIS)** to manage student records. The system maintains information such as student profiles, course registrations, grades, attendance, and fee status.

The university requires the system to provide:

1. A Student Portal where students can view their grades and attendance..
2. A Faculty Portal where instructors can enter grades and attendance records.
3. An Administrative Dashboard where staff can manage student records and generate reports.

Whenever a faculty member updates a student's grade, the change should immediately be reflected in all relevant interfaces. The university also plans to introduce additional dashboards in the future without changing the core student database. **[5+5 (marks)]**

Questions:

- a. How the Student Information System can be implemented using the MVC and PAC architectures. Clearly represent all components graphically.
- b. Which implementation model, MVC or PAC, is more suitable for this system? Justify your answer based on the requirements given in the scenario. .

A4a) MVC Implementation

Model

The Model contains and manages:

- Student records
- Course information
- Grades
- Attendance records
- Fee information

The Model serves as the central data repository and business logic component.

Views

Different Views present the same data to different users:

- Student Portal
- Faculty Portal
- Administrative Dashboard

Each View displays information according to the user's role.

Controllers

Controllers process user actions:

- Students request academic records.
- Faculty members enter grades and attendance.
- Administrators update student information.

PAC Implementation

The system can also be divided into several PAC agents:

Student Agent

- Presentation: Student Portal
- Abstraction: Student-related data
- Control: Communication with other agents

Faculty Agent

- Presentation: Faculty Portal
- Abstraction: Course and grading information
- Control: Coordination with database and other agents

Administration Agent

- Presentation: Administrative Dashboard
- Abstraction: Administrative records
- Control: Communication among agents

The Control components coordinate interactions between the agents.

A4b) MVC is the most suitable architecture.

Justification

1. The system has **one central data source** (student database).
2. Multiple interfaces display the **same underlying information**.
3. Updates to the Model must be reflected in several Views.
4. MVC naturally supports multiple synchronized Views connected to a single Model.
5. The interaction logic is relatively straightforward and does not require autonomous agents.
6. MVC is simpler to implement and maintain for this type of information management system.

Q5) “Equipped with a dictionary, you are perfectly able to check your spelling by reading through the entire paper and correcting mistakes as you spot them. This mundane task is perfectly suited to automation, as the computer can check words against its own list of acceptable spellings”. **[5+5 (marks)]**

Questions:

- a. Considering Spell-checking application, to which usability principle and sub principle this scenario indicates.
- b. According to the identified principle in part a, classify whether the application is Human operated or Computer operated with justification.

A5a) Flexibility----->Task migrability

A5b) It is not desirable, however, to leave this task completely to the discretion of the computer, as most computerized dictionaries do not handle proper names correctly, nor can they distinguish between correct and unintentional duplications of words. In those cases, the task is handed over to the user. The spell-check is best performed in such a cooperative way.